

RAG Literature Review Assistant

Hybrid arXiv search using Qwen keyword prefiltering, Claude HyDE, SBERT + BM25 retrieval, and RRF ranking.

Free queries remaining: 3 / 3 (resets every 1h)

Academic fields

All arXiv Fields ✕ ▼

Restrict to specific arXiv subcategories (optional)

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- ☒ Qwen keyword prefilter
- ☒ Claude justifications

This demo returns the top 5 papers per query.

Search arXiv Literature

Enter your research question

transformers in machine learning ✎

Searches typically take around 2 minutes to complete — please be patient after clicking search.

Find Papers

How the search narrowed down 3M+ papers ↔

Total corpus	After field filter	After keyword filter
3,067,...	3,067,...	1,853,...
	↓ -0%	↓ -40%

Keywords used: transformers , machine learning , deep learning , neural networks , artificial intelligence , natural language processing , language modeling , text generation , tokenization , embedding , attention mechanism , fine-tuning , pretraining , model architecture , data augmentation , cross-lingual translation , transfer learning , unsupervised learning , semi-supervised learning , self-supervised learning , ensemble methods , hyperparameter tuning , evaluation metrics

Top 5 papers

1. Leaner Transformers: More Heads, Less Depth

Why relevant: The paper directly addresses transformer architectures, which are a central topic in modern machine learning. It focuses specifically on redesigning transformer components (multi-head attention and depth), making it highly relevant to a query about transformers in machine learning.

Contribution: Demonstrates that increasing the number of attention heads improves conditioning of the attention block, enabling reduction of model depth by 30-50% in parameter count while maintaining accuracy across vision and language tasks.

Relevance: 8.0/10 · Specificity: 7.0/10

Transformers have reshaped machine learning by utilizing attention mechanisms to capture complex patterns in large datasets, leading to significant improvements in performance. This success has contributed to the belief that "bigger means better", leading to ever-increasing model sizes. This paper challenge this ideology by showing that many existing transformers might be unnecessarily oversized. We discover a theoretical principle that redefines the role of multi-head attention. An important be...

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2. Attention mechanisms in neural networks

Why relevant: The paper directly addresses transformers and attention mechanisms, which are the core components of transformer models in machine learning. It covers transformer variants including autoregressive transformers, bidirectional encoders,

and Vision Transformers, making it highly relevant to the query.

Contribution: A comprehensive mathematical treatment of attention mechanisms and transformer architectures covering theory, applications across NLP and vision, scaling laws, and current limitations.

Relevance: 9.0/10 · Specificity: 7.0/10

Attention mechanisms represent a fundamental paradigm shift in neural network architectures, enabling models to selectively focus on relevant portions of input sequences through learned weighting functions. This monograph provides a comprehensive and rigorous mathematical treatment of attention mechanisms, encompassing their theoretical foundations, computational properties, and practical implementations in contemporary deep learning systems. Applications in natural language processing, computer...

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3. Machine learning of microstructure--property relationships in materials leveraging microstructure representation from foundational vision transformers

Year: 2025

RRF score: 0.0164

Why relevant: The paper directly uses transformer architectures (vision transformers) as a core component of its methodology. However, the primary focus is on materials science applications rather than advancing transformer methods themselves.

Contribution: Proposes using pre-trained foundational vision transformers (CLIP, DINOv2, SAM) as task-agnostic feature extractors for machine learning of microstructure-property relationships in materials science.

Relevance: 5.0/10 · Specificity: 7.0/10

Machine learning of microstructure--property relationships from data is an emerging approach in computational materials science. Most existing machine learning efforts focus on the development of task-specific models for each microstructure--property relationship. We propose utilizing pre-trained foundational vision transformers for the extraction of task-agnostic microstructure features and subsequent light-weight machine learning of a microstructure-dependent property. We demonstrate our appro...

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4. Long-Short Transformer: Efficient Transformers for Language and Vision

Year: 2021

RRF score: 0.0161

Why relevant: The paper is directly about transformer architectures in machine learning, specifically improving the self-attention mechanism. It addresses core transformer components and demonstrates applications across both language and vision domains.

Contribution: Proposes Long-Short Transformer (Transformer-LS), an efficient self-attention mechanism with linear complexity that combines long-range attention with dynamic projection and short-term local attention for both language and vision tasks.

Relevance: 8.0/10 · Specificity: 8.0/10

Transformers have achieved success in both language and vision domains. However, it is prohibitively expensive to scale them to long sequences such as long documents or high-resolution images, because self-attention mechanism has quadratic time and memory complexities with respect to the input sequence length. In this paper, we propose Long-Short Transformer (Transformer-LS), an efficient self-attention mechanism for modeling long sequences with linear complexity for both language and vision tas...

[View on arXiv →](#)

5. Energy-Efficient Deep Reinforcement Learning with Spiking Transformers

Year: 2025

RRF score: 0.0161

Why relevant: The paper directly involves transformer architectures in a machine learning context, specifically adapting transformer structures with spiking neural networks for reinforcement learning tasks. While transformers are central to the work, the primary focus is on energy efficiency and reinforcement learning rather than transformers themselves.

Contribution: Proposes a Spike-Transformer Reinforcement Learning (STRL) algorithm combining spiking neural networks with transformer attention mechanisms for energy-efficient reinforcement learning.

Relevance: 6.0/10 · Specificity: 8.0/10

Agent-based Transformers have been widely adopted in recent reinforcement learning advances due to their demonstrated ability to solve complex tasks. However, the high computational complexity of Transformers often results in significant energy consumption, limiting their deployment in real-world autonomous systems. Spiking neural networks (SNNs), with their biologically inspired structure, offer an

energy-efficient alternative for machine learning. In this paper, a novel Spike-Transformer Reinf...

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